General Information



Simon Dorrer

KV VLSI Design, 336.051, WiSe 26/27

JOHANNES KEPLER UNIVERSITY LINZ

INSTITUTE FOR INTEGRATED CIRCUITS & QUANTUM COMPUTING

JOHANNES KEPLER UNIVERSITÄT LINZ
Altenberger Straße 69
4040 Linz, Österreich
jku.at

Introduction



About Me – Simon Dorrer

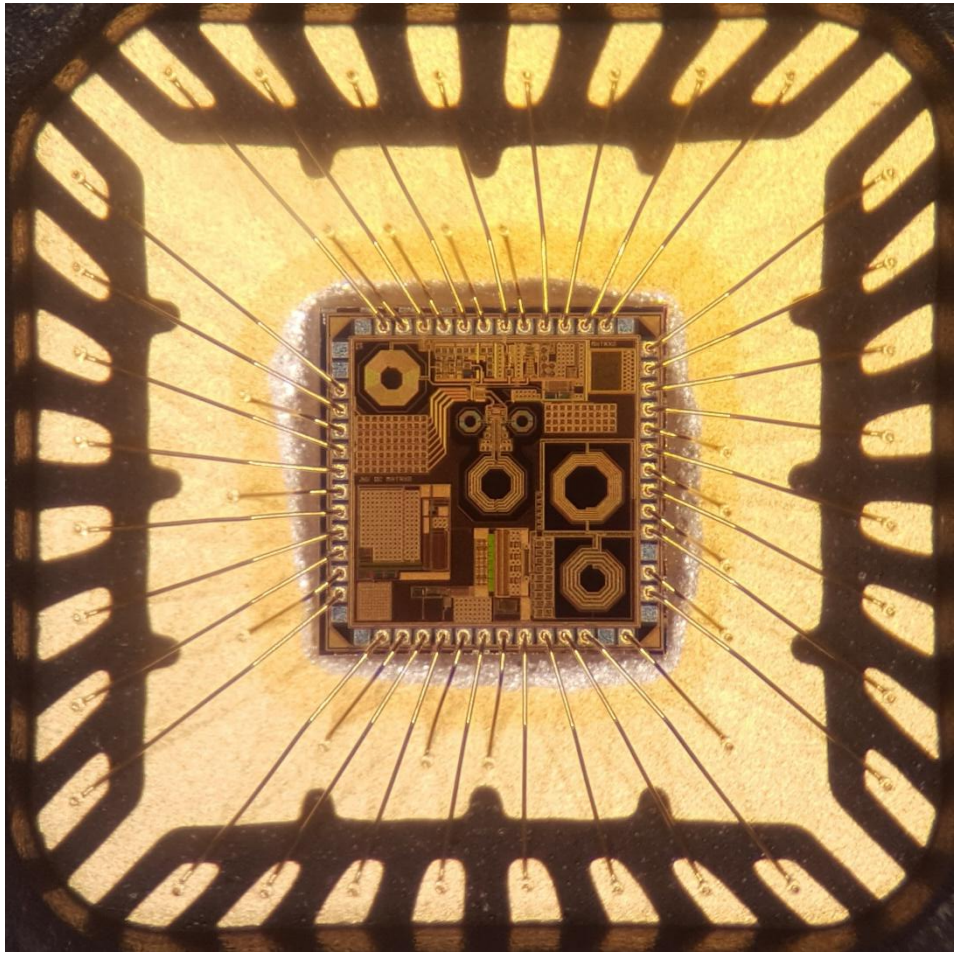
- PhD student @ JKU
- Research and teaching assistant @ IICQC
- Open-source projects:
  - Agentic AI AMS and RF design flows
  - mm-wave six-port receivers
  - TinyWhisper SoC
  - Event-based ADCs
  - Tutorials and templates
- Email: simon.dorrer@jku.at



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Institute for Integrated Circuits and Quantum Computing

- Departments:
  - Integrated Circuits (Prof. Pretl)
  - Quantum Computing (Prof. Kueng)
- Design of analog, digital, and RF integrated circuits (e.g. for smartphones, Internet of Things, 6G, medical electronics, ADCs, ...)
- Design, verification, and testing of circuits and systems
- Open-source tool development for chip design https://github.com/iic-jku/iic-osic-tools
- Agentic AI chip design

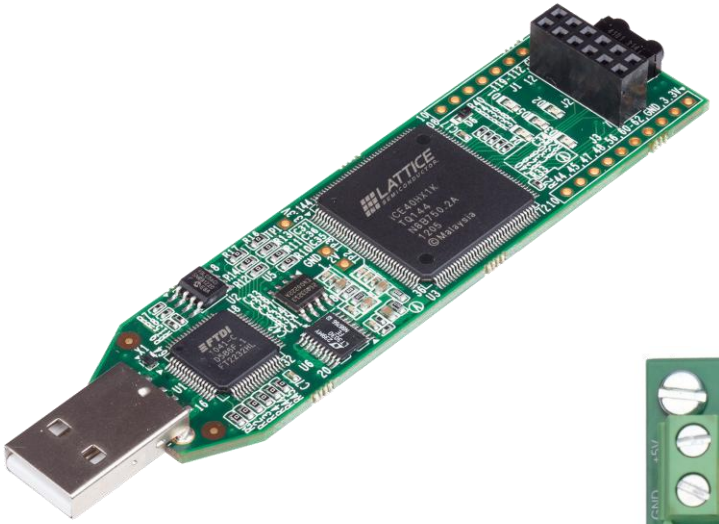


Raise your hand if:

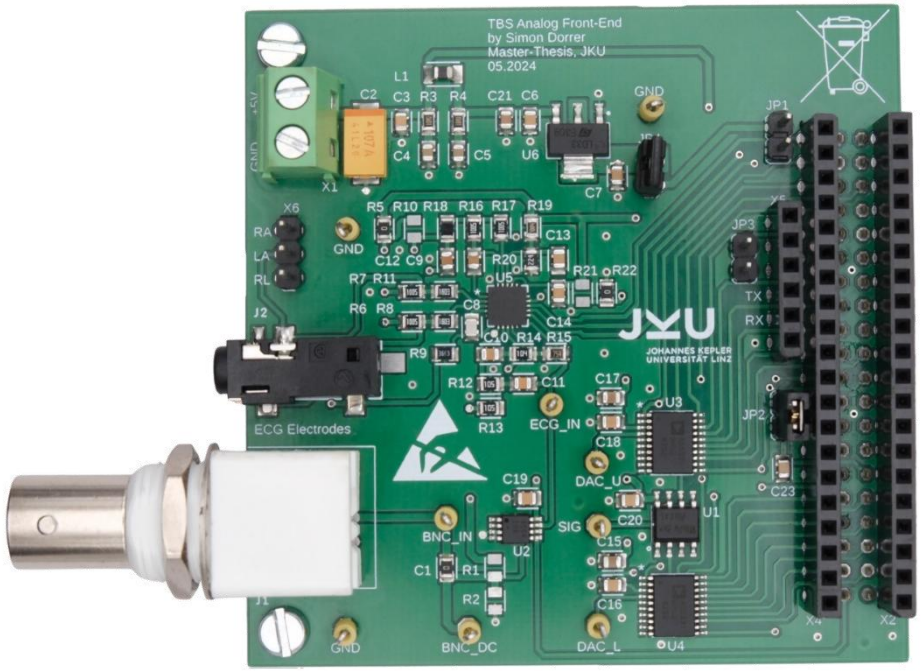
- You know what **VLSI** stands for
- You know what **ASIC** stands for
- You know what **HDL** stands for
- You have already used any HDL before
- You have already worked with **MCU**s or **FPGA**s
- You have already designed an **ASIC**
- You want to learn how to build a **tapeout**-ready chip
- You regularly use **AI agents**

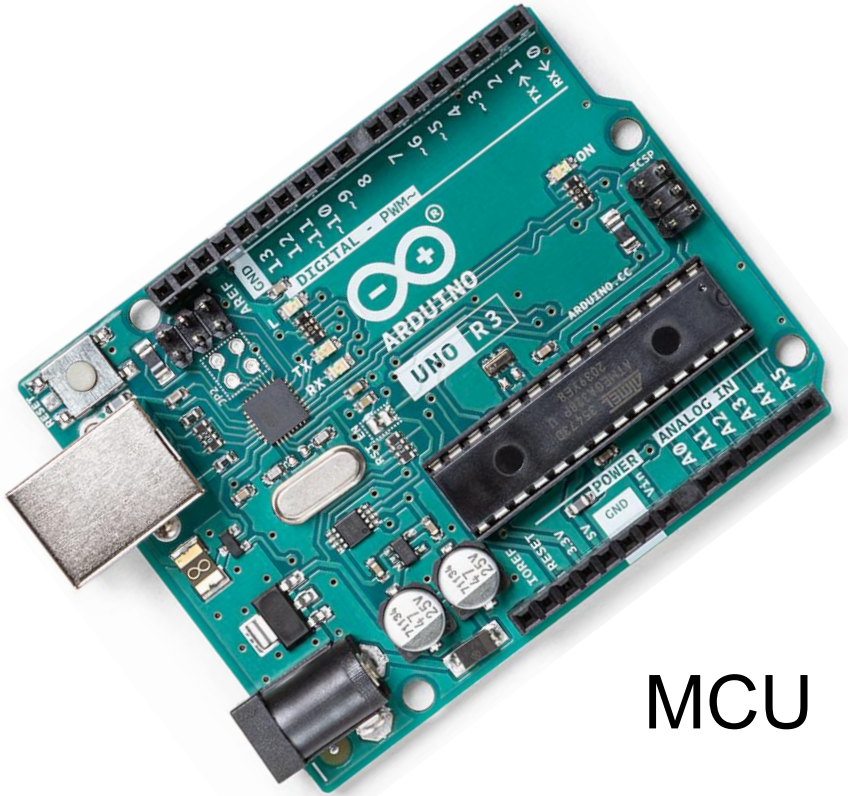


What are electronic systems?



FPGA



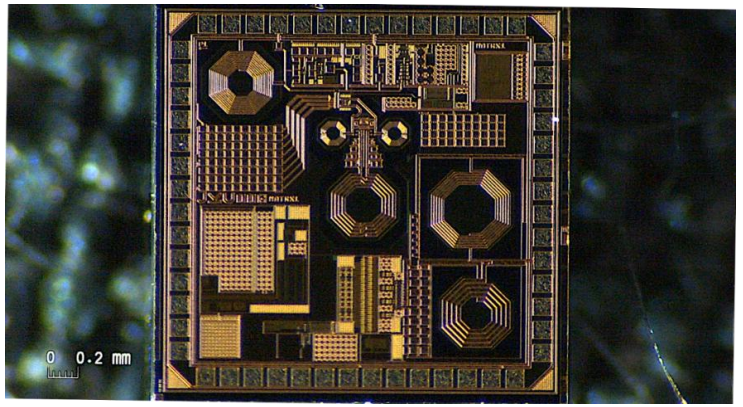


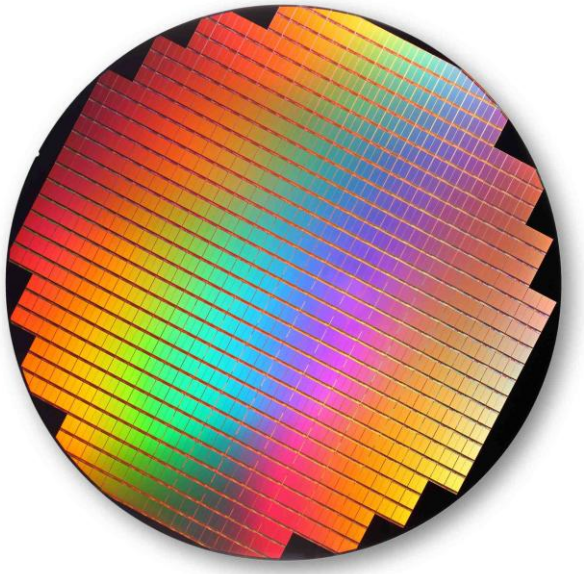
MCU

Mixed-Signal Designs

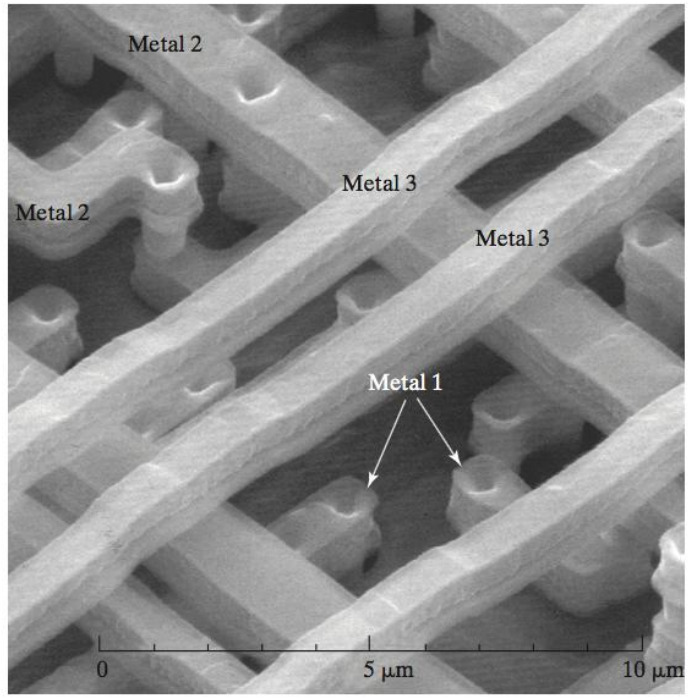
What are integrated circuits?

- Electrical circuits that are processed on silicon wafers.
- Components and interconnections are built up layer by layer through various processing steps.









https://www.allaboutcircuits.com/news/diamonds-vie-to-replace-silicon-as-next-semiconductor-material/

From Sand to Silicon: The Making of a Microchip



Overview & Goals



Requirements

- Bachelor CS / Master ELIT course (elective subject)
- Basic understanding of CMOS transistors
- Good understanding of digital logic (gates, flip-flops, etc.)
- Basic experience with a hardware description language (e.g., VHDL)
- Own laptop and ≥ 20 GB of free disk space (VSCode, Git, IIC-OSIC-TOOLS)
- Optional:
  - KV Analog Circuit Design: https://github.com/iic-jku/analog-circuit-design
  - KV Design of Complex Integrated Circuits: https://github.com/iic-jku/design-complex-ic

Content



- KV VLSI Design: https://github.com/iic-jku/vlsi-design
- Introduction
- From VHDL to SystemVerilog & modeling guidelines
- Functional verification with cocotb
- Logic synthesis, standard-cell library & equivalence checking
- Sequential elements & static timing analysis (STA)
- Synthesis optimizations & low-power design
- Clock domain crossing (CDC) & design for test (DFT)
- RTL-to-GDS flow with LibreLane, chip assembly & padframe
- Project: tapeout-ready RISC-V SoC

Grading

- **Part 1 – Theory (50%):**
  - Theoretical lectures mixed with small examples
  - Written exam before Christmas (**17.12.**)
- **Part 2 – Project (50%):**
  - Implementation of a full tapeout-ready RISC-V SoC

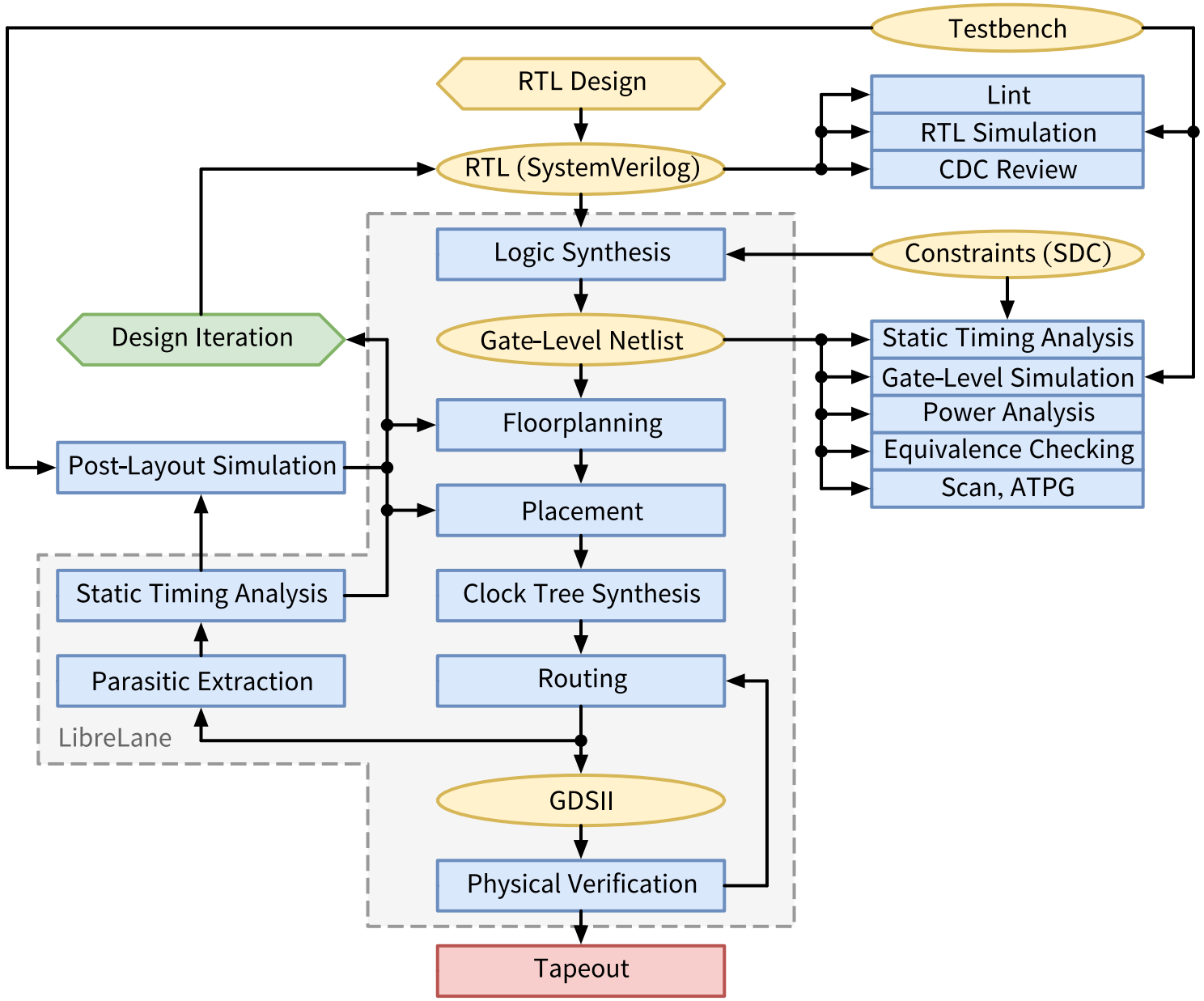
    + own extension (analog or digital)
  - Oral presentation
  - **The best project might be taped out!**
  - **Possible extension to seminar work (+3ECTS) or thesis!**

Goals? After this course you should:

- Be able to handle the IIC-OSIC-TOOLS
- Be able to handle GitHub
- Be able to read and write SystemVerilog RTL code
- Be able to verify RTL with self-checking cocotb testbenches
- Be able to understand and explain RTL-to-GDS flows
- Be able to handle timing (STA), testability (DFT), and clock domains
- Be able to build tapeout-ready ASICs with open-source tools
- Be able to use LLM agents efficiently (optional)

RTL to GDS Flow

- **Ovals**: design data
- **Boxes**: steps
- **Dashed**: LibreLane



“The aha moment of seeing your circuit come to life will stay long after the equations are forgotten.” – Peter Kinget

Analog Mixed-Signal Design Flow & Template



GitHub Template

- **Solderpad Hardware License v2.1**
- Use this template or `git clone …`
- `make all`
- **Tapeout-ready AMS chip!**
- **Tutorial** based on Quarto
- **Regression tests** via GitHub actions
- **Unbelievable?**
  - **Try it out now! →**
- Tapeout proven?

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An Open-Source Analog Mixed-Signal Chip Design Template & Tutorial for the ihp-sg13cmos5l Open-PDK

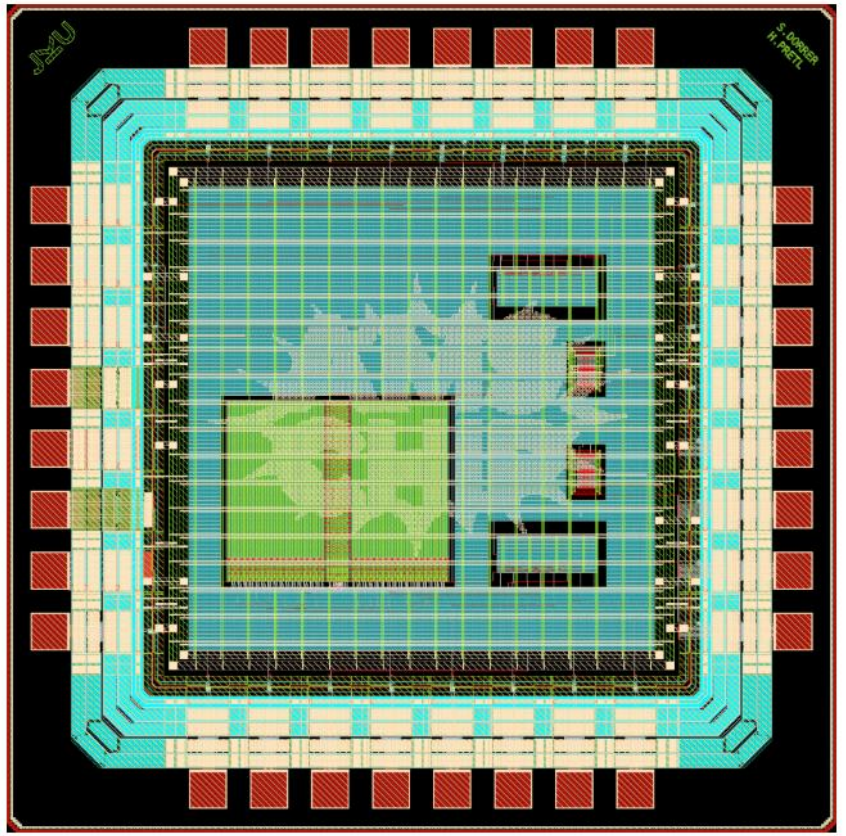
License Solderpad Hardware License v2.1 | quarto-publish passing | regression passing | License Check passing | Tutorial online

DOI 10.5281/zenodo.22115134

(c) 2026 Simon Dorrer and Harald Pretl

Institute for Integrated Circuits and Quantum Computing, Johannes Kepler University (JKU), Linz, Austria

> **Important**
>
> This repository requires the IIC-OSIC-TOOLS container with tag `2026.09` or later.



Chip render of the ihp-sg13cmos5l analog mixed-signal template chip (1.6 mm x 1.6 mm).

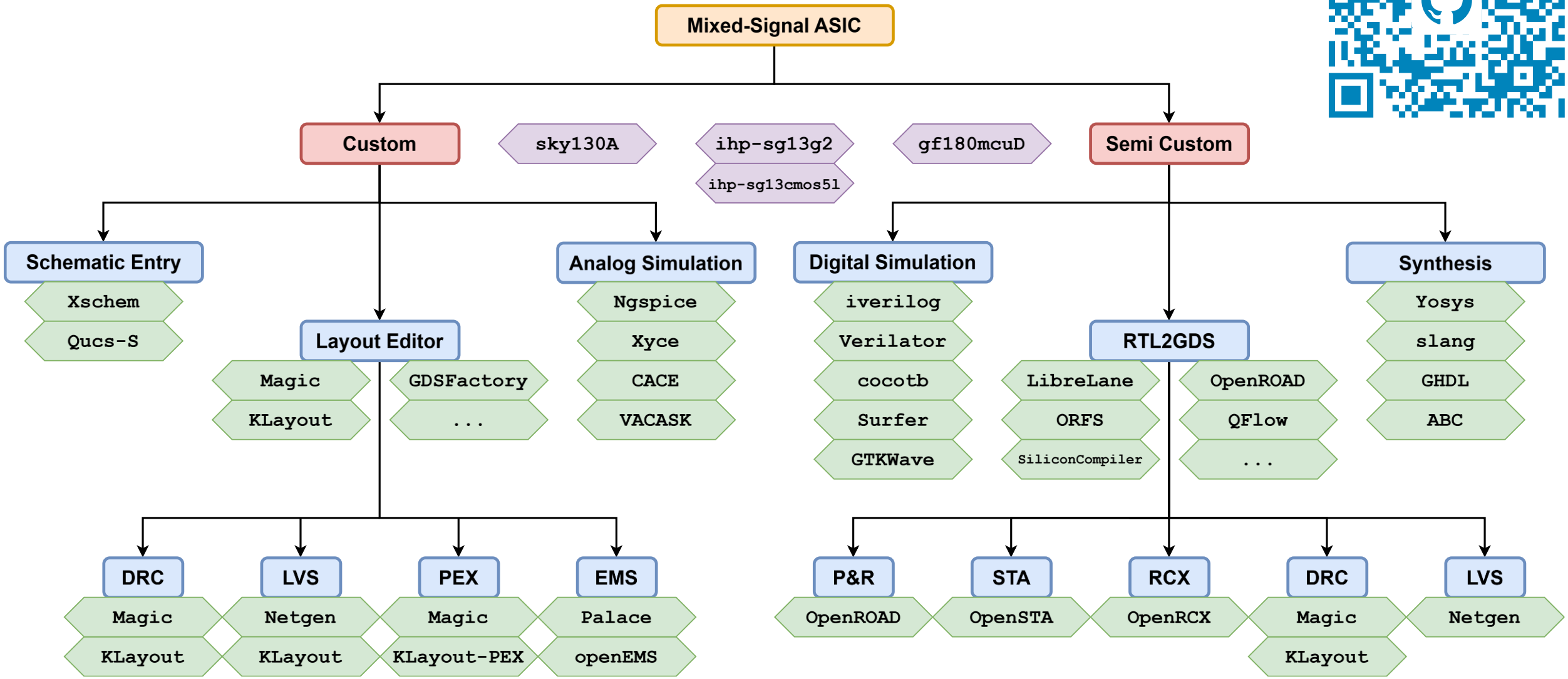
Flow Overview



- Simulation, linting, and hardening of RTL
- Layout checks and reports of hardened digital macro
- Simulation of all analog testbenches (incl. PVT and MC)
- Layout checks of analog & digital blocks (DRC, LVS, and PEX)
- Digital-on-top top-level assembly and layout check (DRC)
  - Seal ring and pad frame generation
  - Macro placement and wiring also with non-default rules (NDRs)
  - Logo and fill insertion
- Full top-level LVS and simulation without black boxing
- Bonding diagram

IIC-OSIC-TOOLS

IIC-OSIC-TOOLS
OPEN-SOURCE IC DESIGN · JKU LINZ



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Open-Source vs. Commercial (1/2)

Flow step	Open source	Commercial (examples)	Main gap
Lint	Verilator	VC SpyGlass, Questa Lint	No testability rules, naming rules only in Verible
Simulation	Icarus Verilog, Verilator	Xcelium, VCS, Questa One Sim	Verilator: two-state, no SDF. Icarus: no timing checks
CDC	OpenCDC	Questa CDC, VC SpyGlass CDC	Under development: false errors, missed crossings
Synthesis	Yosys, ABC	Design Compiler, Genus	Larger area, higher power
Equivalence	EQY	Conformal, Formality	Language coverage of the Yosys front ends
Scan and ATPG	OpenROAD, Fault, Difetto	TestMAX, Tessent	Stuck-at faults only. Difetto: early stage, not in the image

Open-Source vs. Commercial (2/2)

Flow step	Open source	Commercial (examples)	Main gap
Place and route	OpenROAD	Innovus, Fusion Compiler	Larger area, higher power, longer run times
Extraction	OpenRCX	Quantus, StarRC	Rule-based, needs calibrated rules
STA	OpenSTA	PrimeTime, Tempus	No crosstalk delay in image 2026.10
Power	OpenSTA	PrimePower, Voltus, Joules	Average only: no peak power, no RTL estimate
IR drop, EM	PDNSim	Voltus	Static IR drop, no EM limits
DRC, LVS	Magic, KLayout, Netgen	Calibre, Pegasus, IC Validator	Commercial PDKs: rule decks usually for commercial tools only
Power intent	`read_upf`	Conformal LP, VC LP	UPF subset, no open checker known

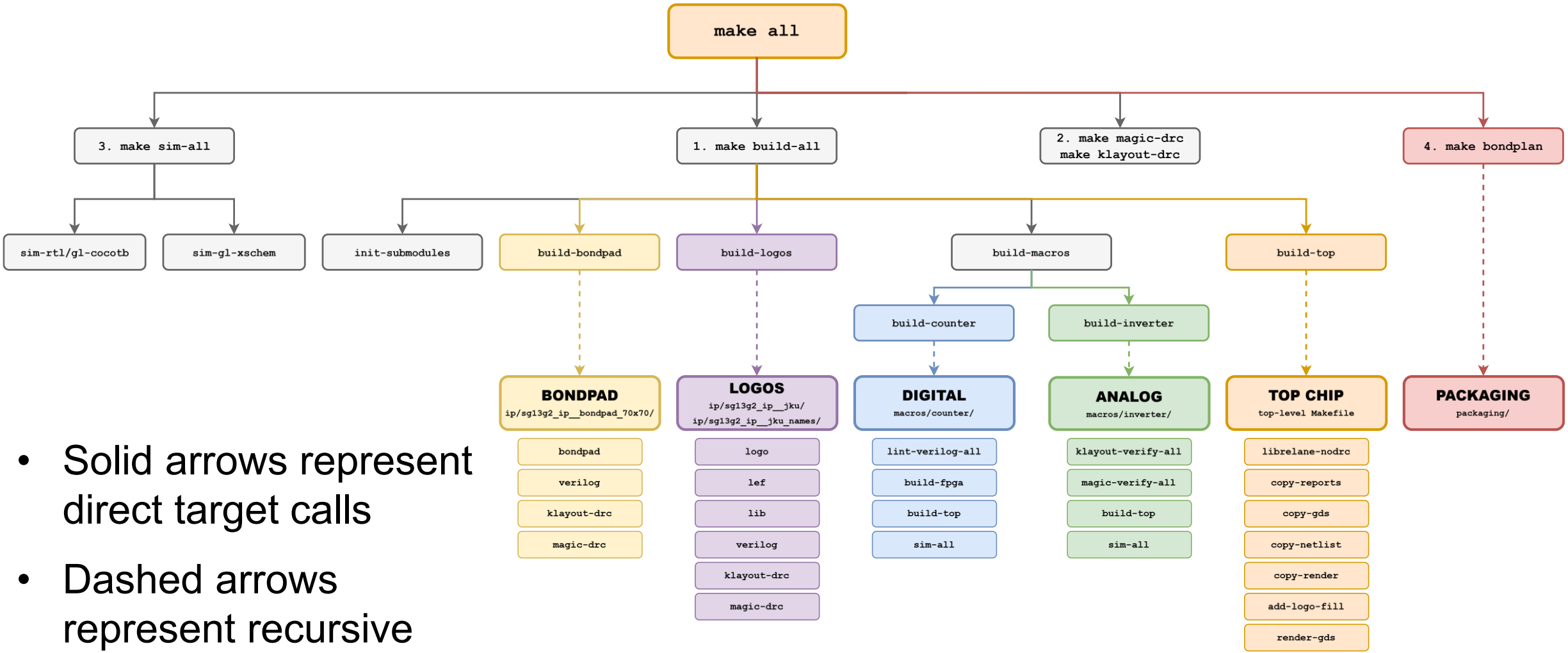
Example: FSM at 50 MHz in IHP 130 nm, Yosys and OpenROAD against Genus and Innovus: **53 % more area, 118 % more power**

Folder Structure

- Structure discussed with FOSSi community
- Easily share and reproduce results of other macros



Makefile Structure

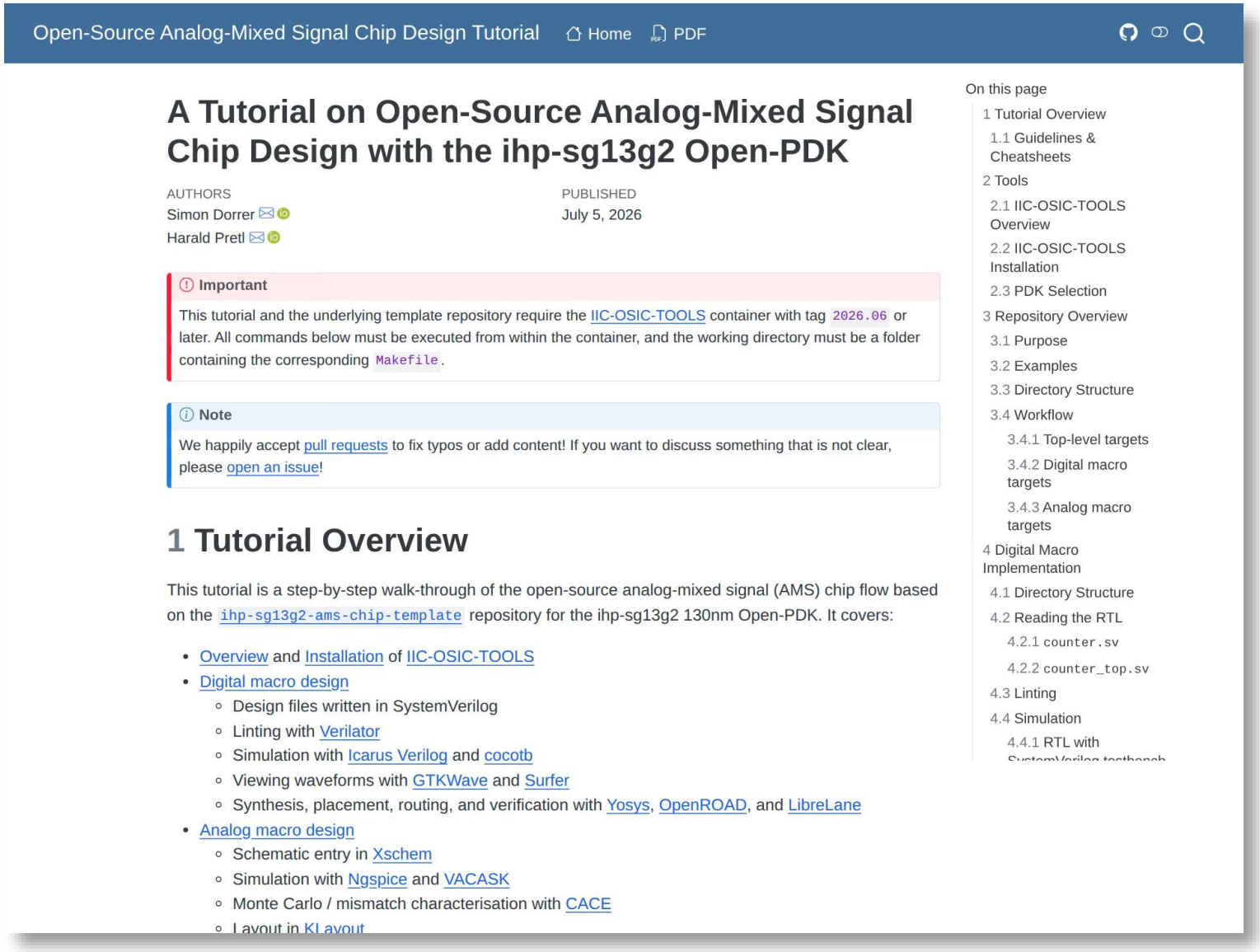


- Solid arrows represent direct target calls
- Dashed arrows represent recursive target calls

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Tutorial

- Walk through all IC design steps to get started
- Using IHP's 130 nm CMOS process
- Simple exercises to learn by doing



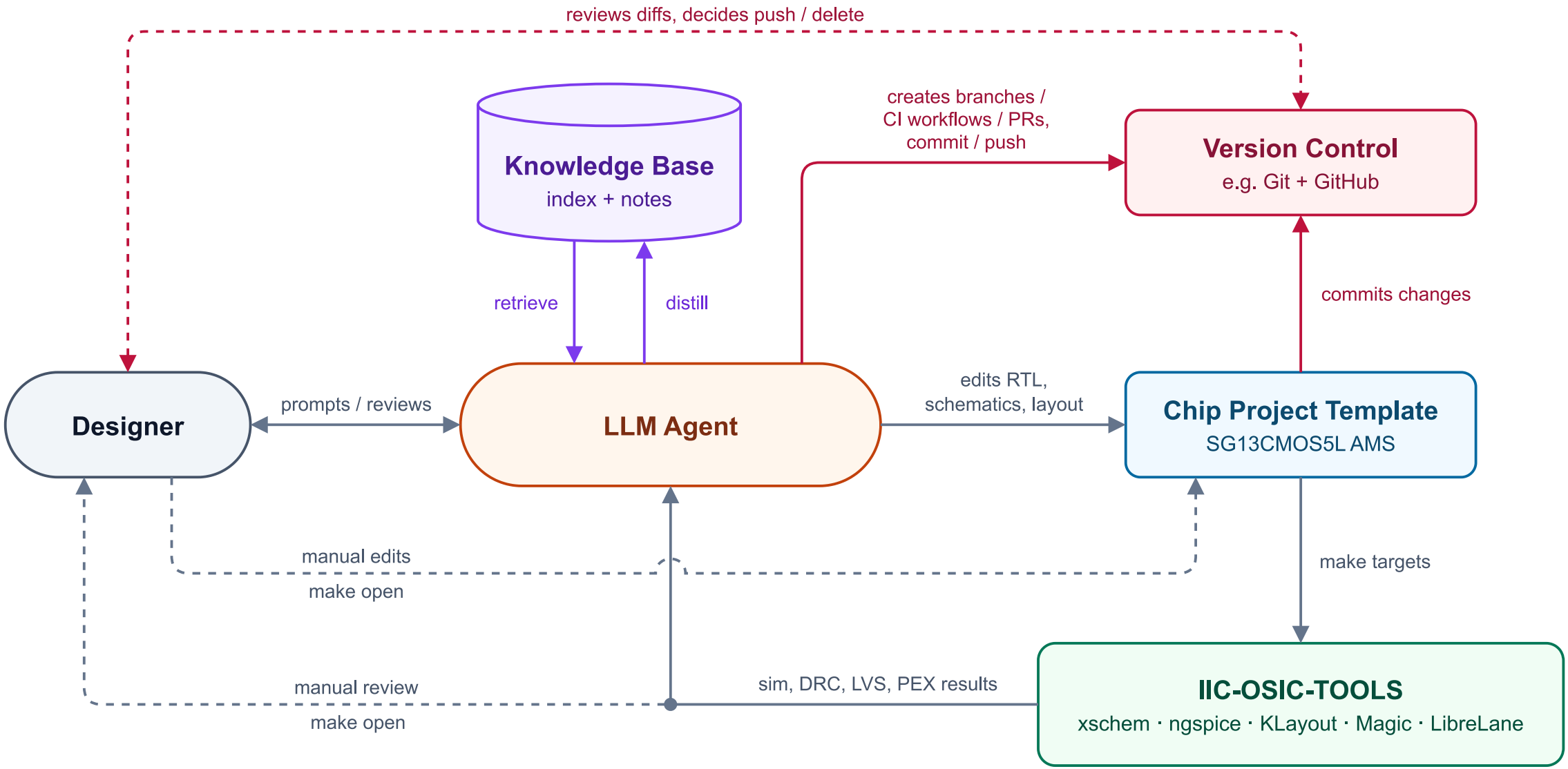
Agentic AI Chip Design



Are you allowed to use AI?

- **Yes,** as a **learning partner**, not as a substitute for your own thinking!
- **Written exam:** no aids, only what you have mastered counts
- **Oral presentation:** you have to explain what you did and why
- **First** try to understand and solve a problem on your own
- **Then** use AI to check your solution, see alternatives, or explain a topic
- **Goal:** the competence to get there yourself, not the fastest answer
- **In the end, you want to learn something. Do not cheat yourself!**
- **We always need people who understand how to prompt and review the results.**

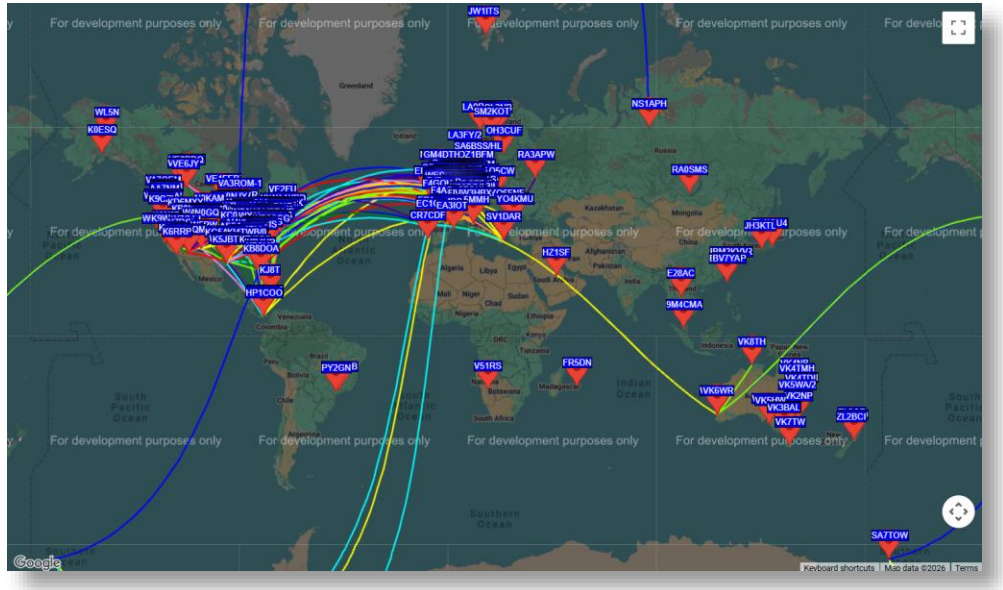
Workflow


reviews diffs, decides push / delete
Knowledge Base
index + notes
creates branches /
CI workflows / PRs,
commit / push
Version Control
e.g. Git + GitHub
retrieve
distill
commits changes
Designer
prompts / reviews
LLM Agent
edits RTL,
schematics, layout
Chip Project Template
SG13CMOS5L AMS
manual edits
make open
make targets
manual review
make open
sim, DRC, LVS, PEX results
IIC-OSIC-TOOLS
xschem · ngspice · KLayout · Magic · LibreLane

TinyWhisper SoC

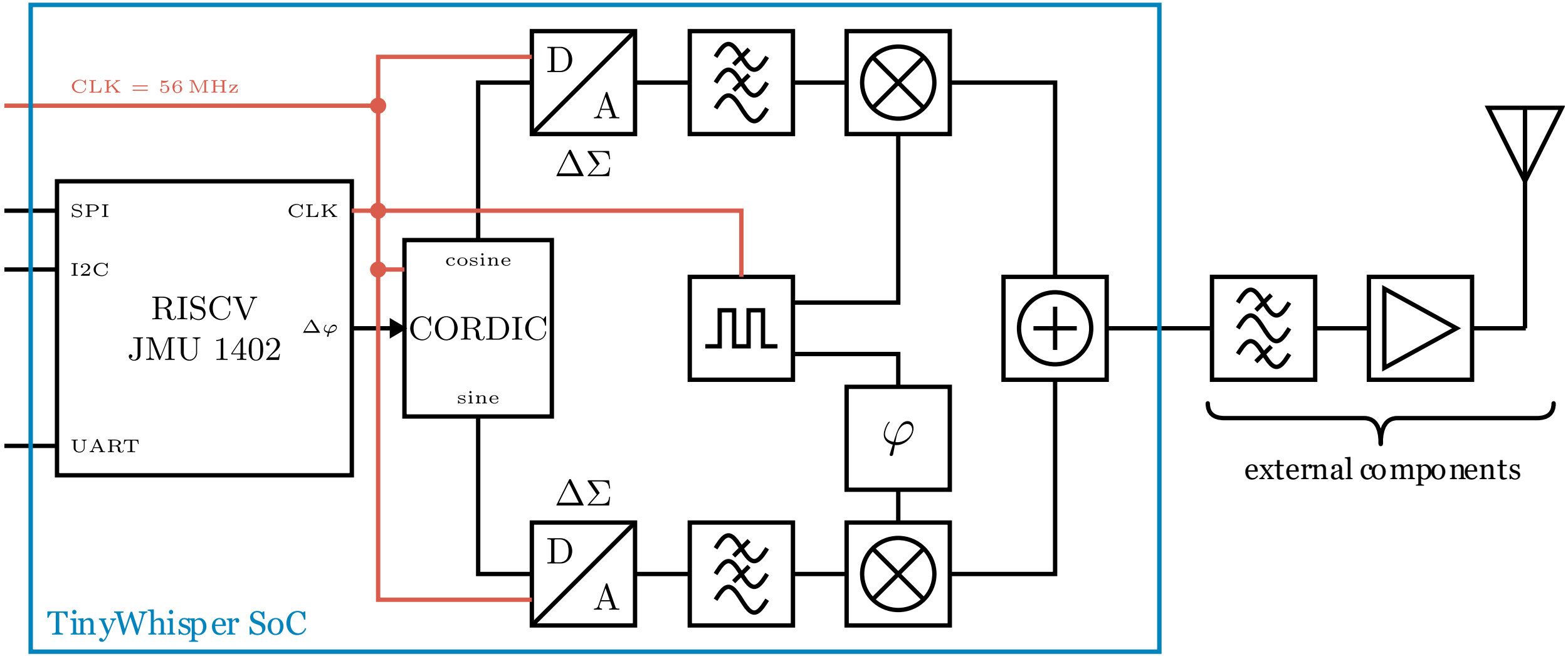


TinyWhisper SoC



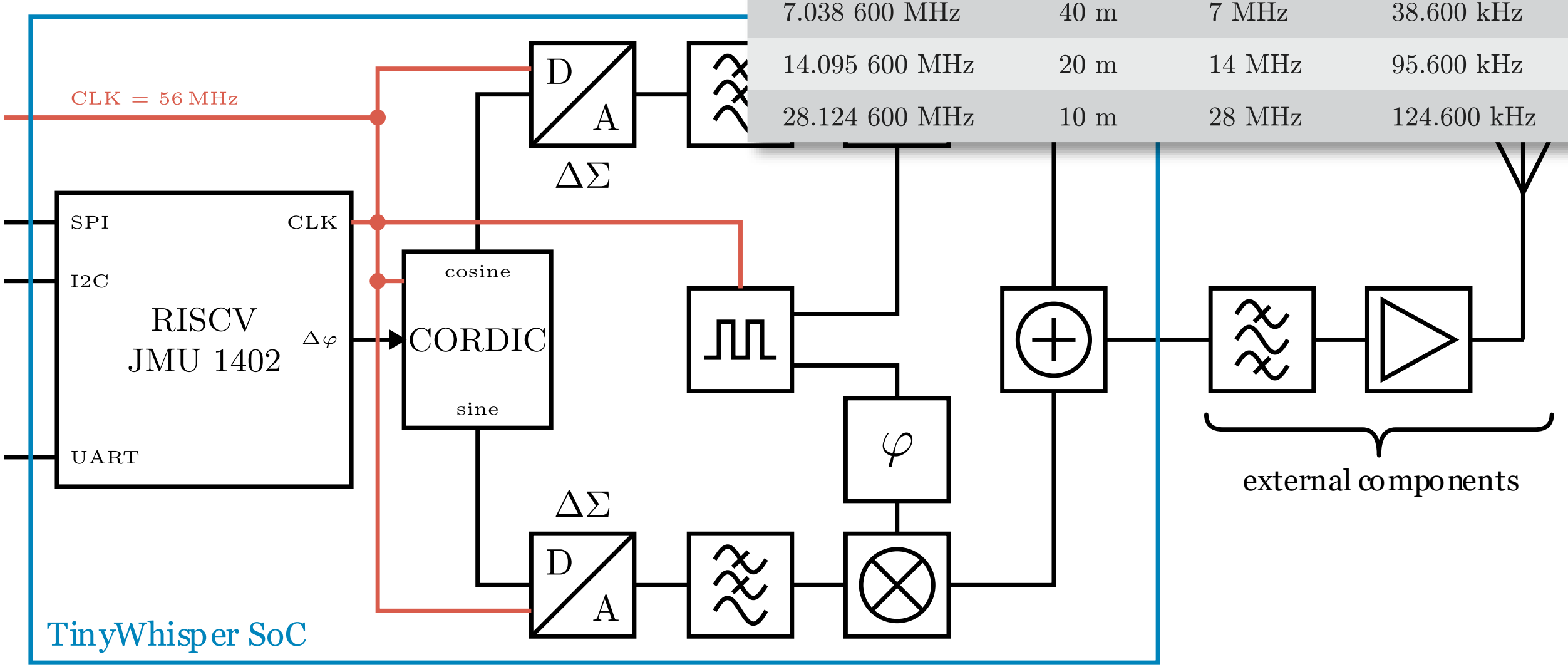
- Cooperation: JMU Würzburg & JKU Linz
- Weak Signal Propagation Reporter (WSPR):
  - Popular digital mode in ham radio
  - Propagation studies @ 80 m to 10 m bands
  - Transmits short message (callsign, location, and power)
  - Receivers report to central database via Internet
  - WSPRnet: https://www.wsprnet.org/drupal/wsprnet/map
  - 4-FSK modulation (Δf = 1.46 Hz, T = 0.683 s)
  - SNR down to -31 dB @ 2500 Hz
  - MODEM (coding, framing, symbol mapping) in FW (C) on RISC-V CPU

Block Diagram



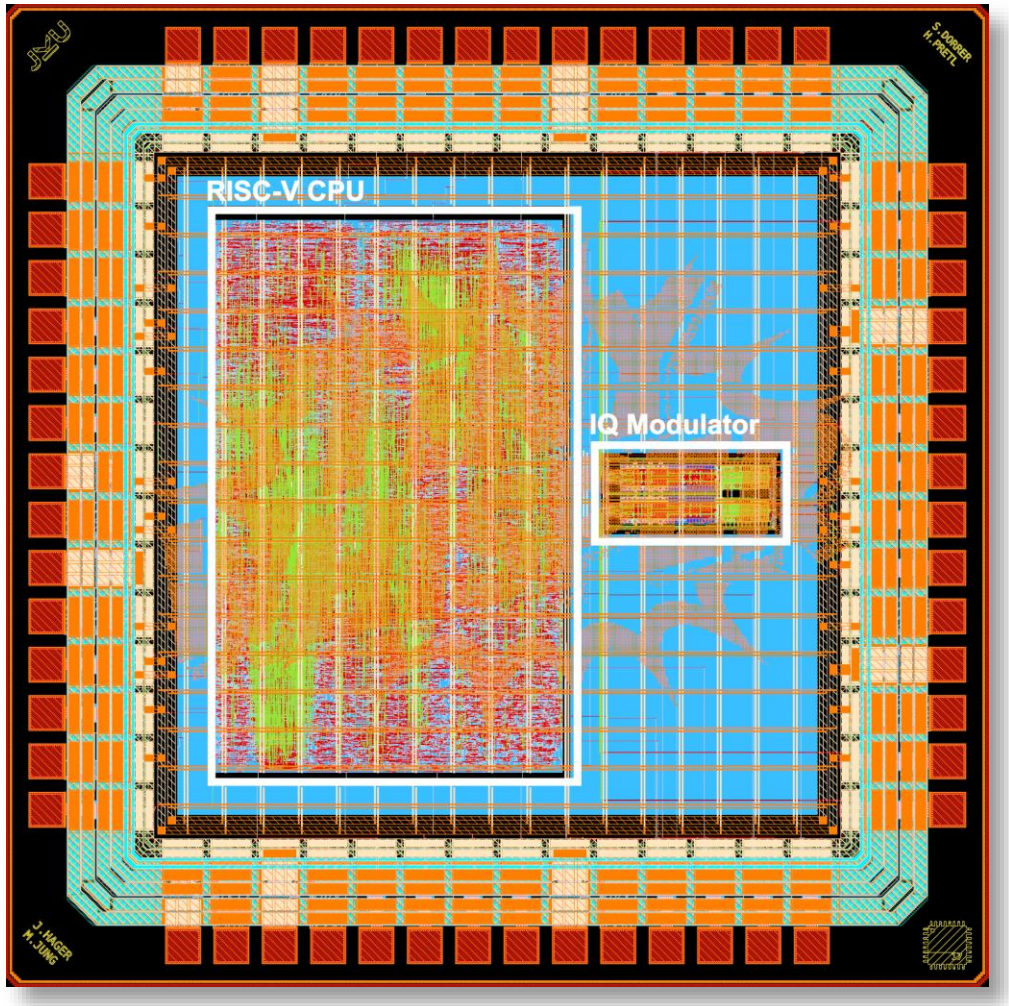
Block Diagram

RF	Band	LO	IF
3.592 600 MHz	80 m	3.5 MHz	92.600 kHz
7.038 600 MHz	40 m	7 MHz	38.600 kHz
14.095 600 MHz	20 m	14 MHz	95.600 kHz
28.124 600 MHz	10 m	28 MHz	124.600 kHz



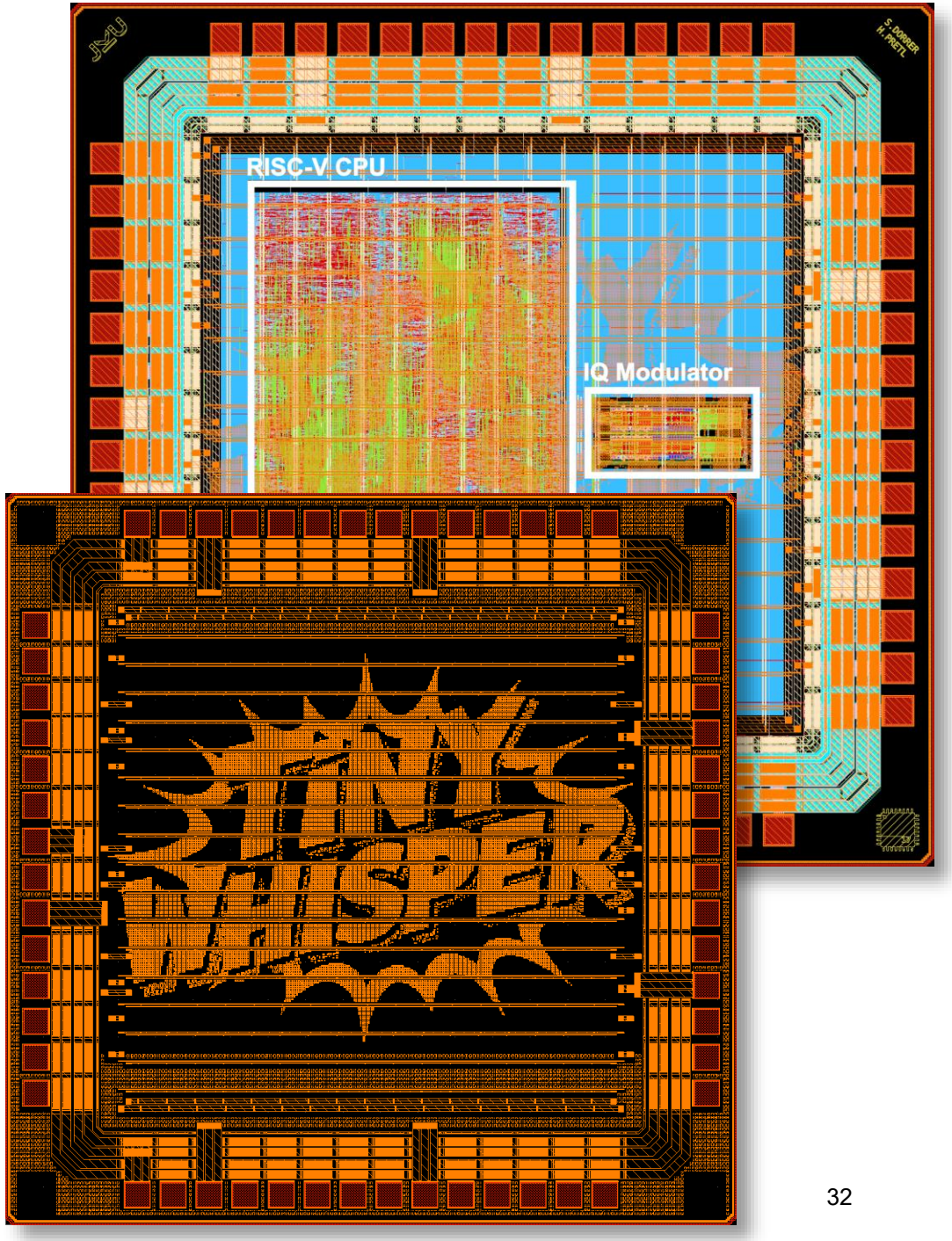
Overview

- 130 nm IHP SG13CMOS Open-PDK
- 2000 µm × 2000 µm (4 mm²)
- 1270 µm × 1270 µm (1.61 mm²)



Overview

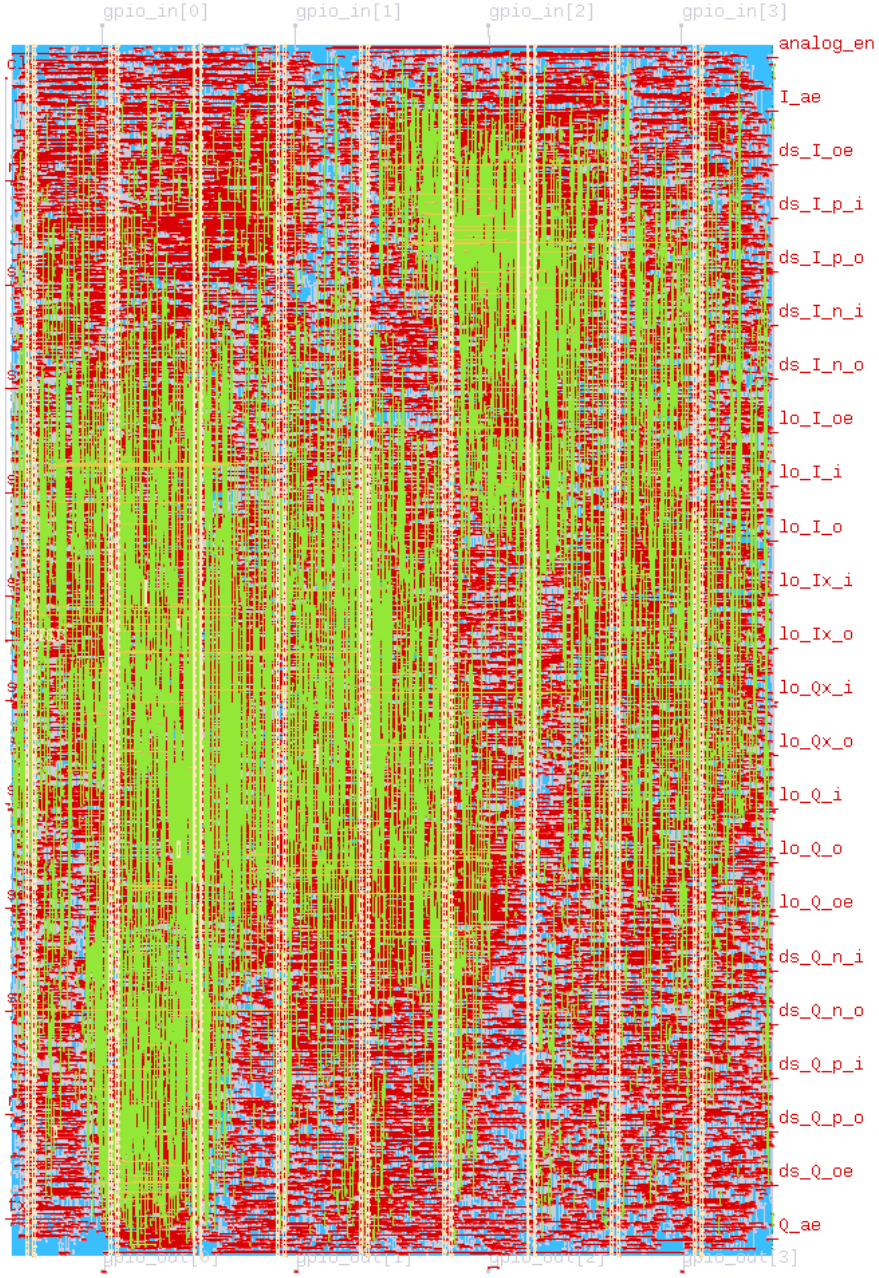
- 130 nm IHP SG13CMOS Open-PDK
- 2000 µm × 2000 µm (4 mm²)
- 1270 µm × 1270 µm (1.61 mm²)
- Logo with ArtistIC by PULP platform
- QFN-48 package
- 56 pads (8 bonded to package ground)
- 1.5 V supply voltage
- 3.3 V I/O supply voltage
- **Chip in production!**



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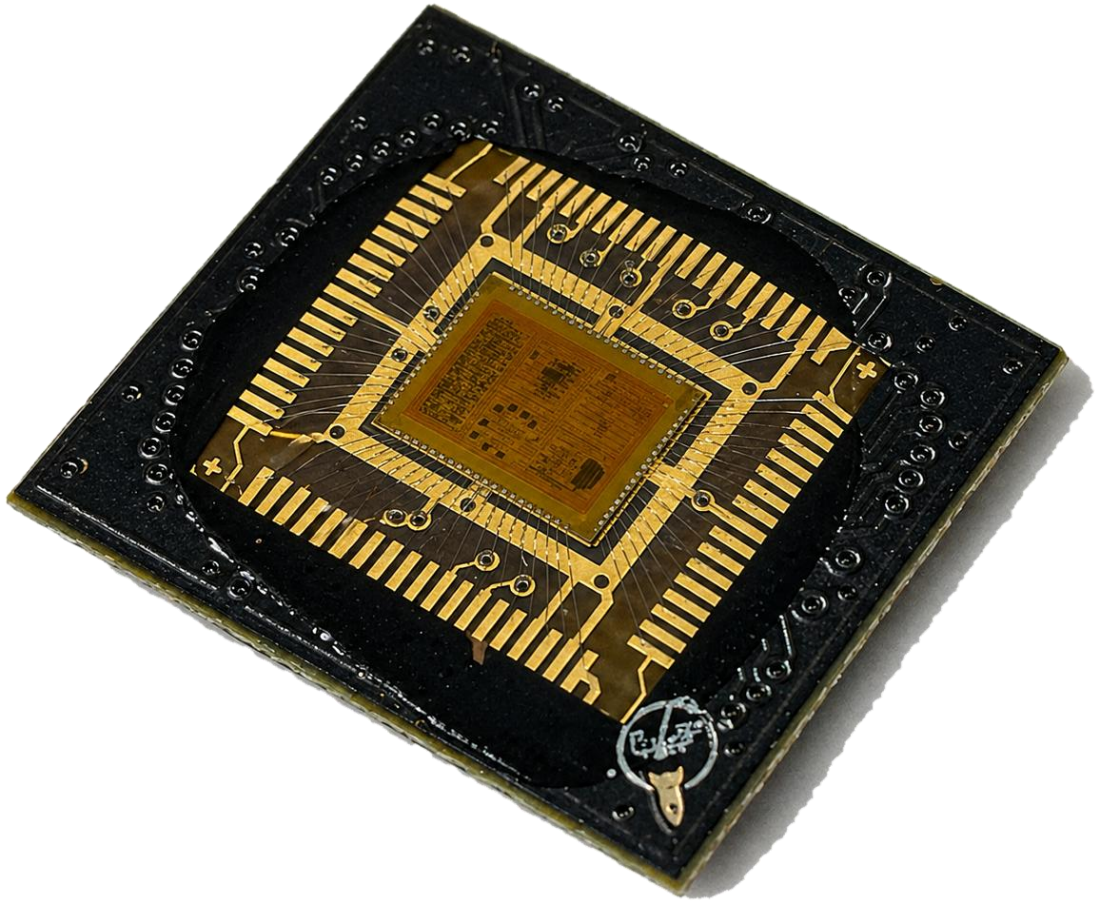
RISC-V CPU

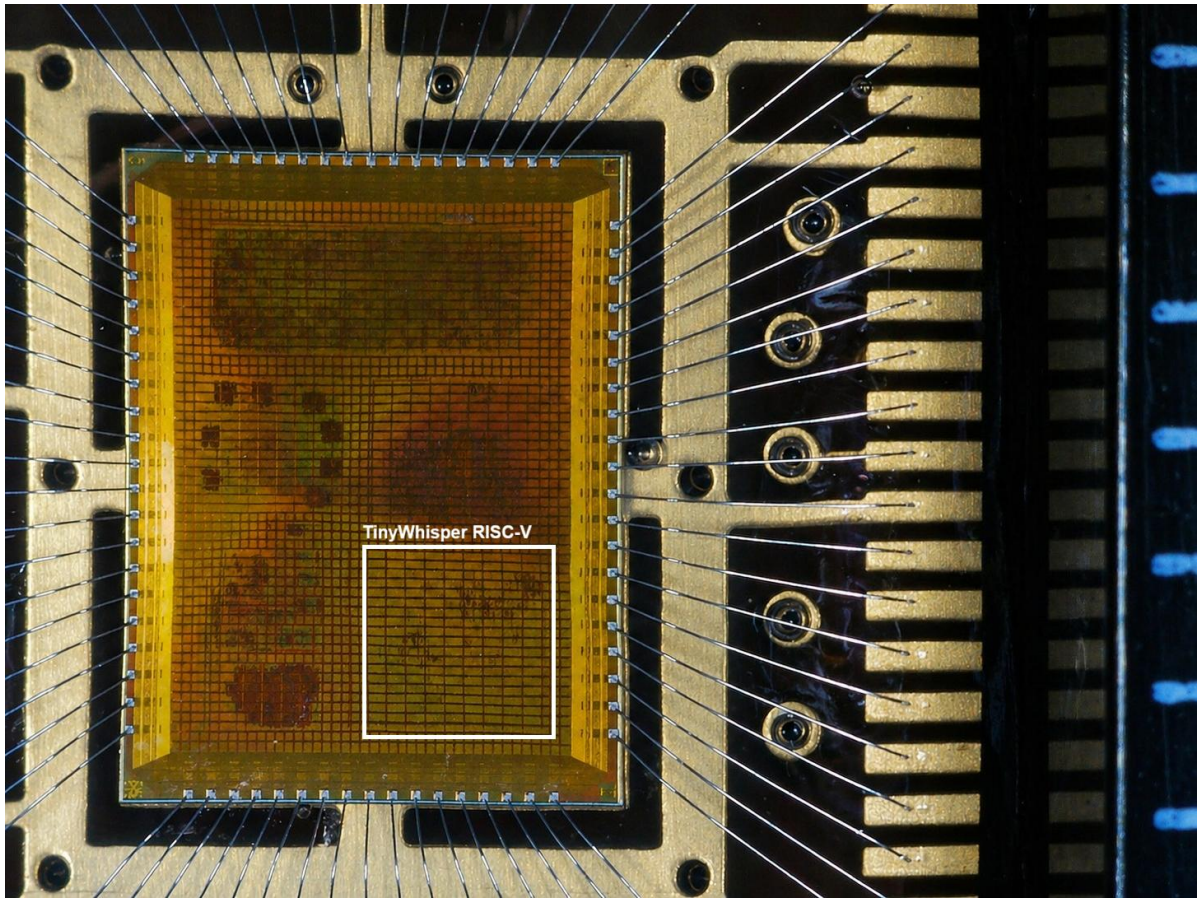
- RISC-V *rv32e_zicsr* embedded CPU
  - Multi-cycle, no pipelining
  - 56 MHz clock frequency
  - SPI, I2C, UART, GPIO are memory-mapped I/O
- Baseband signal generation:
  - 30-bit iterative CORDIC ($\Delta f = 0.05$ Hz)
  - Single-bit DSM (2nd order, OSR ≤ 256)
  - Configurable via software
- 25 % duty cycle LO generation for mixer



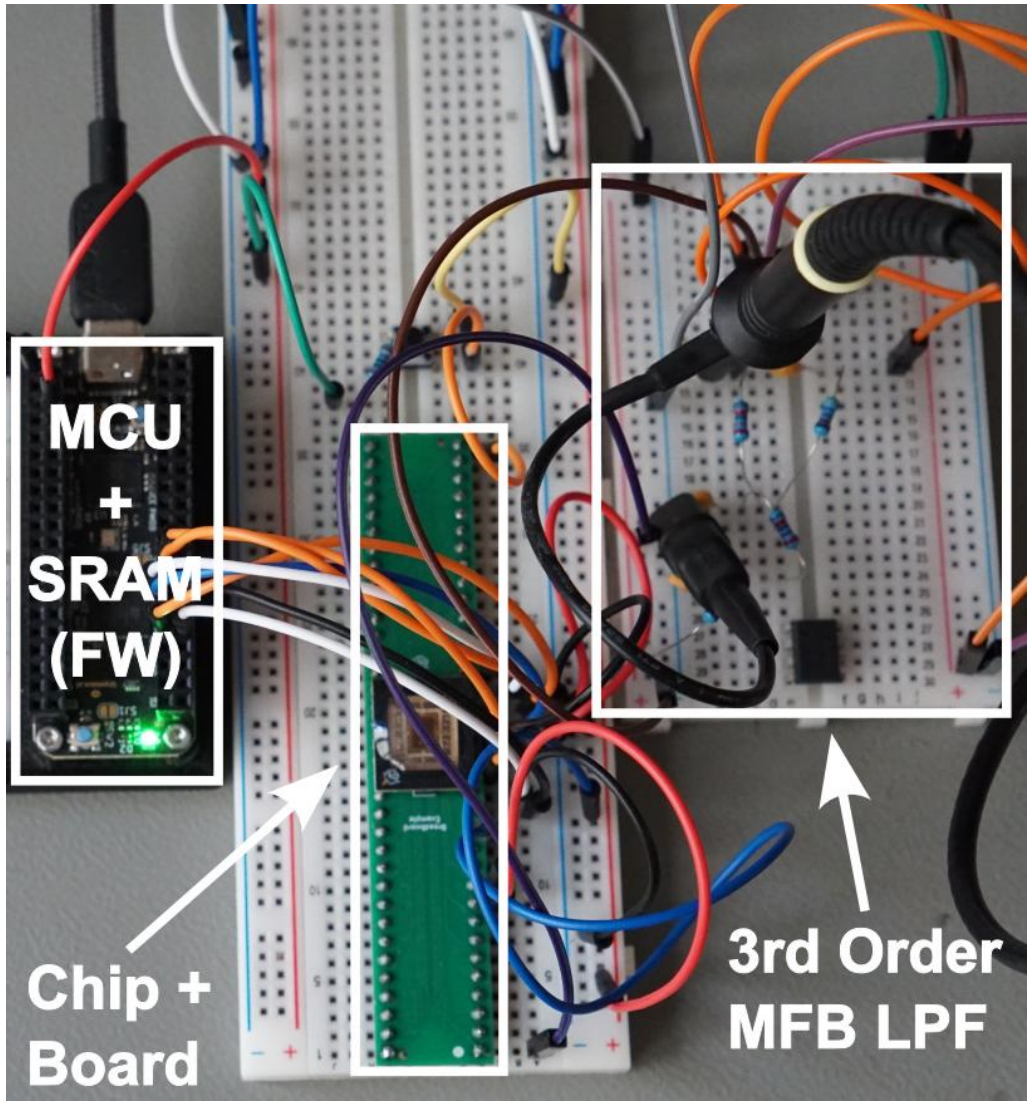
RISC-V CPU – wafer.space Run 1

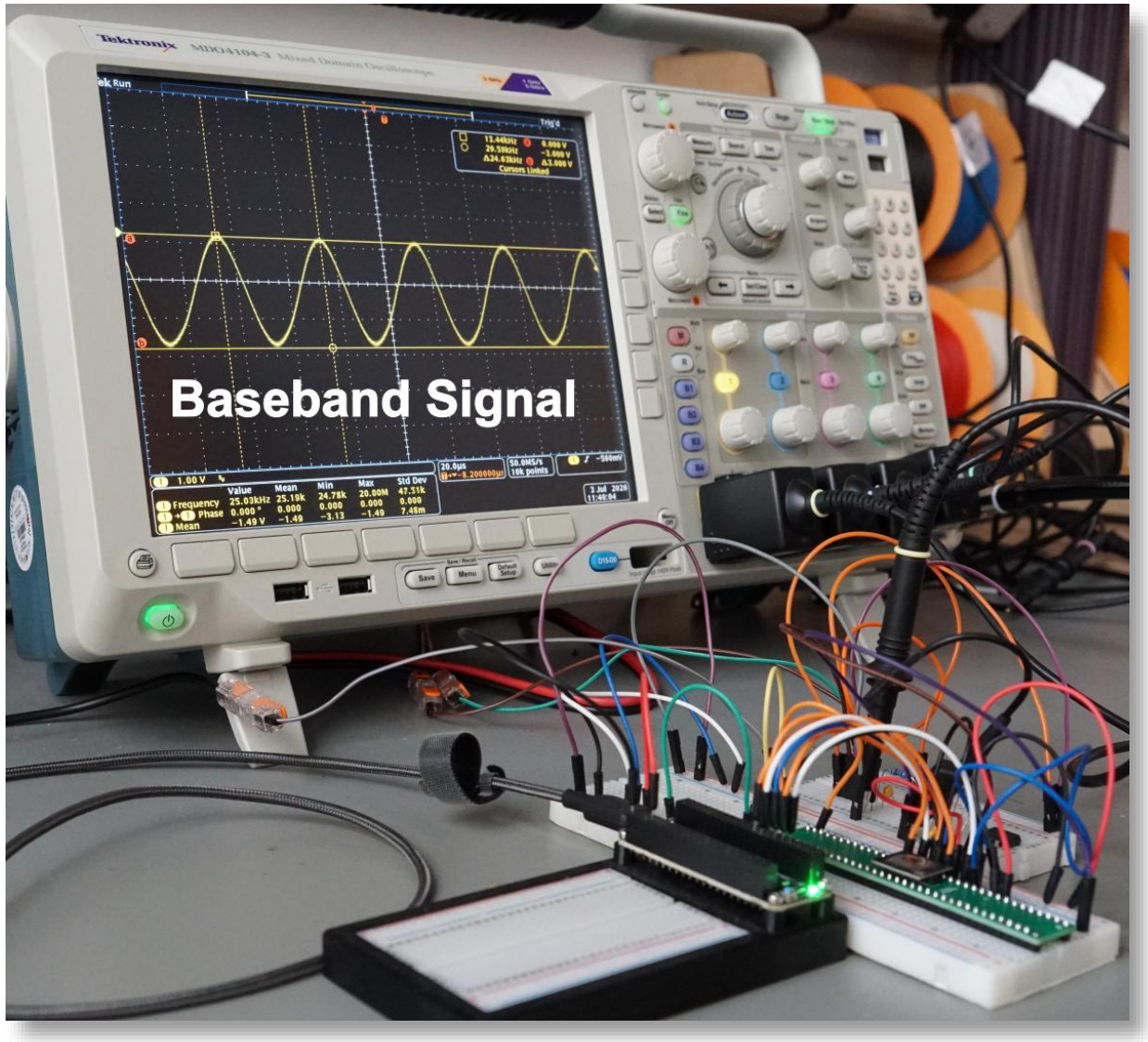
- GlobalFoundries GF180MCU
- Multi-project student tapeout (digital filters, mini games, …)





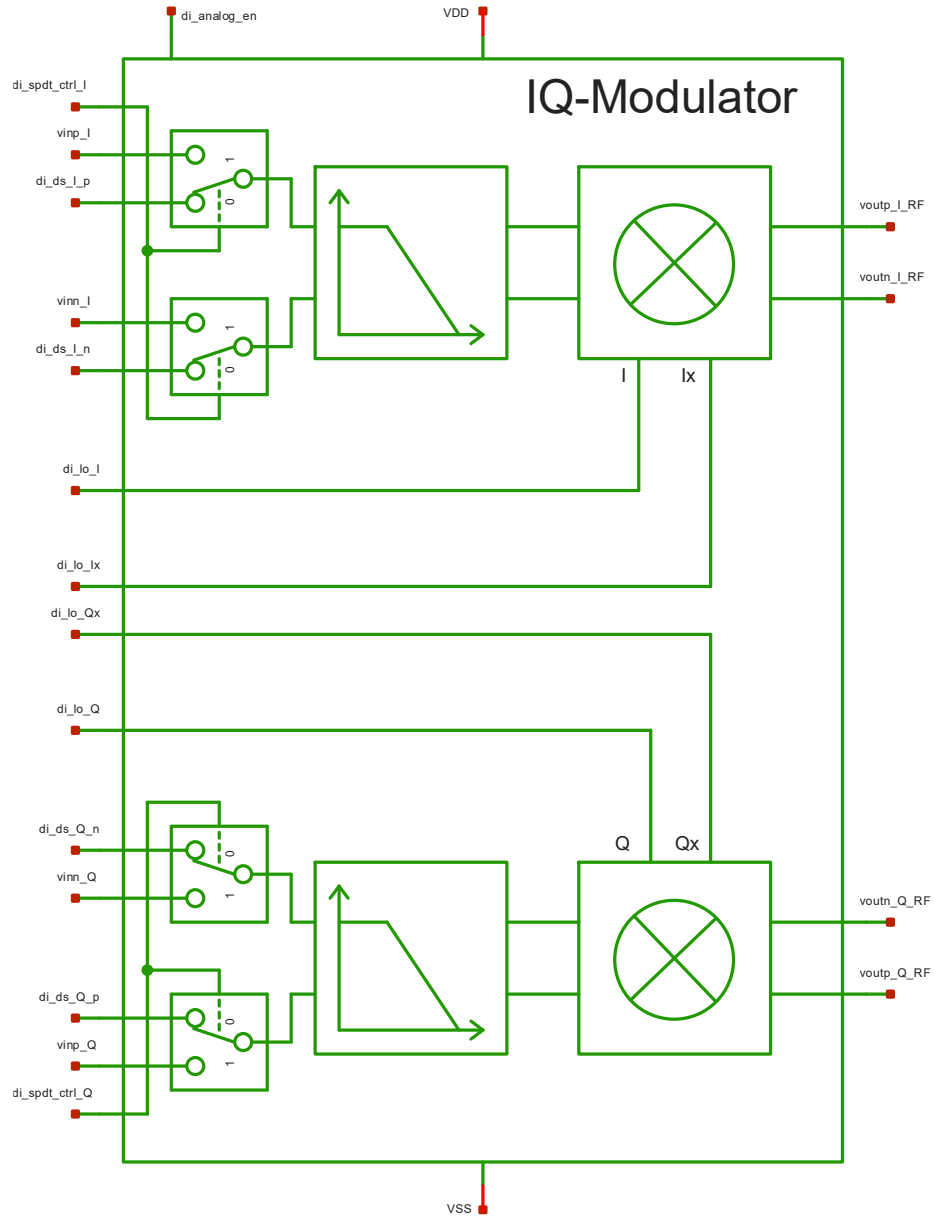
RISC-V CPU – wafer.space Run 1



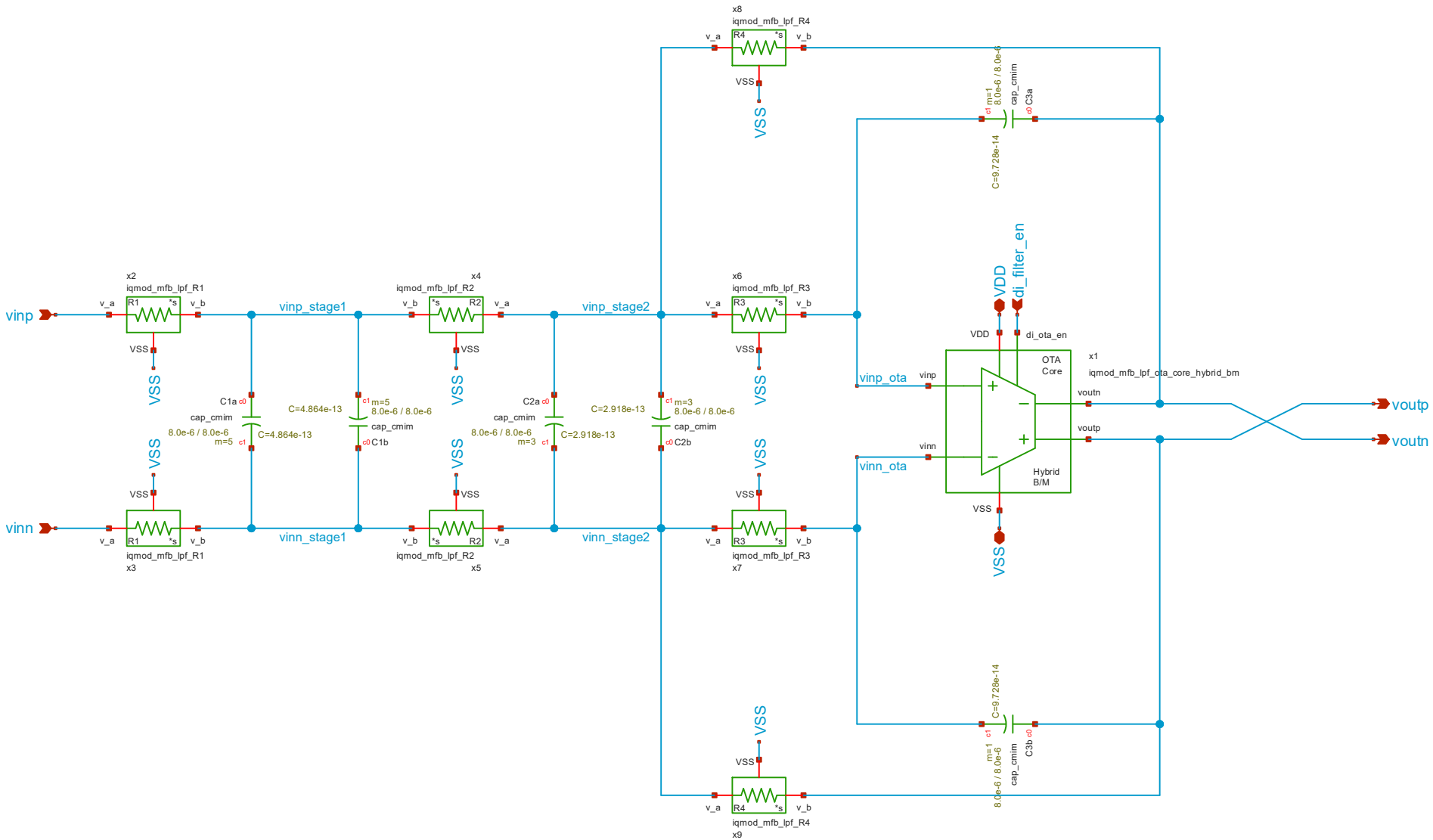


IQ Modulator

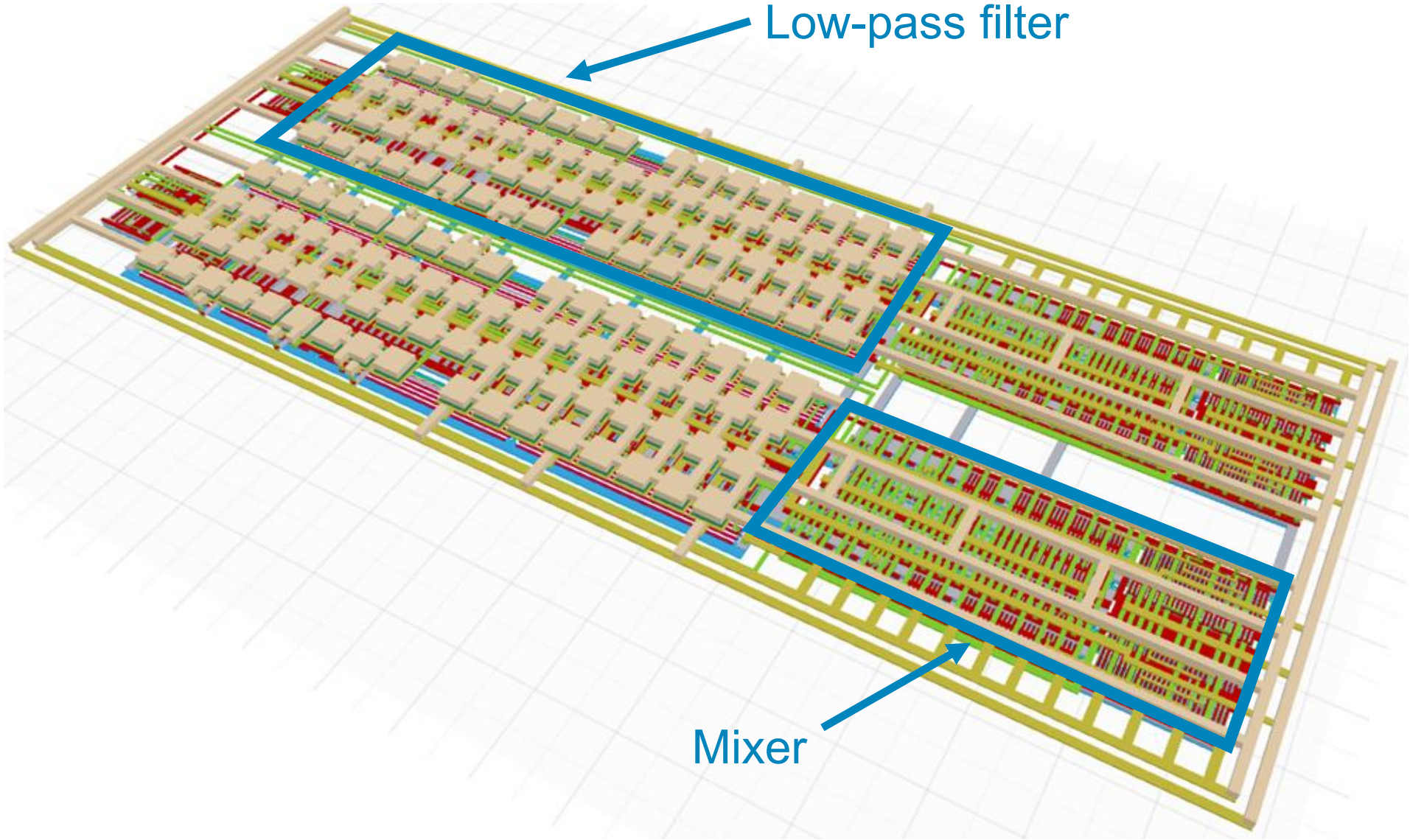
- 3rd order multiple-feedback low-pass filter
  - Butterworth
  - 400 kHz cutoff frequency
  - Barthélemy / Manfredini inverter-based OTA
- Passive CMOS voltage-mode mixer
  - Simple transmission-gate (TG) switch
  - Single-to-differential conversion circuit
- Only inverter- and TG-based circuits
- IQ-TX: Not only WSPR, but FT8, SSB, CW, …



Schematic Drawing is Art!

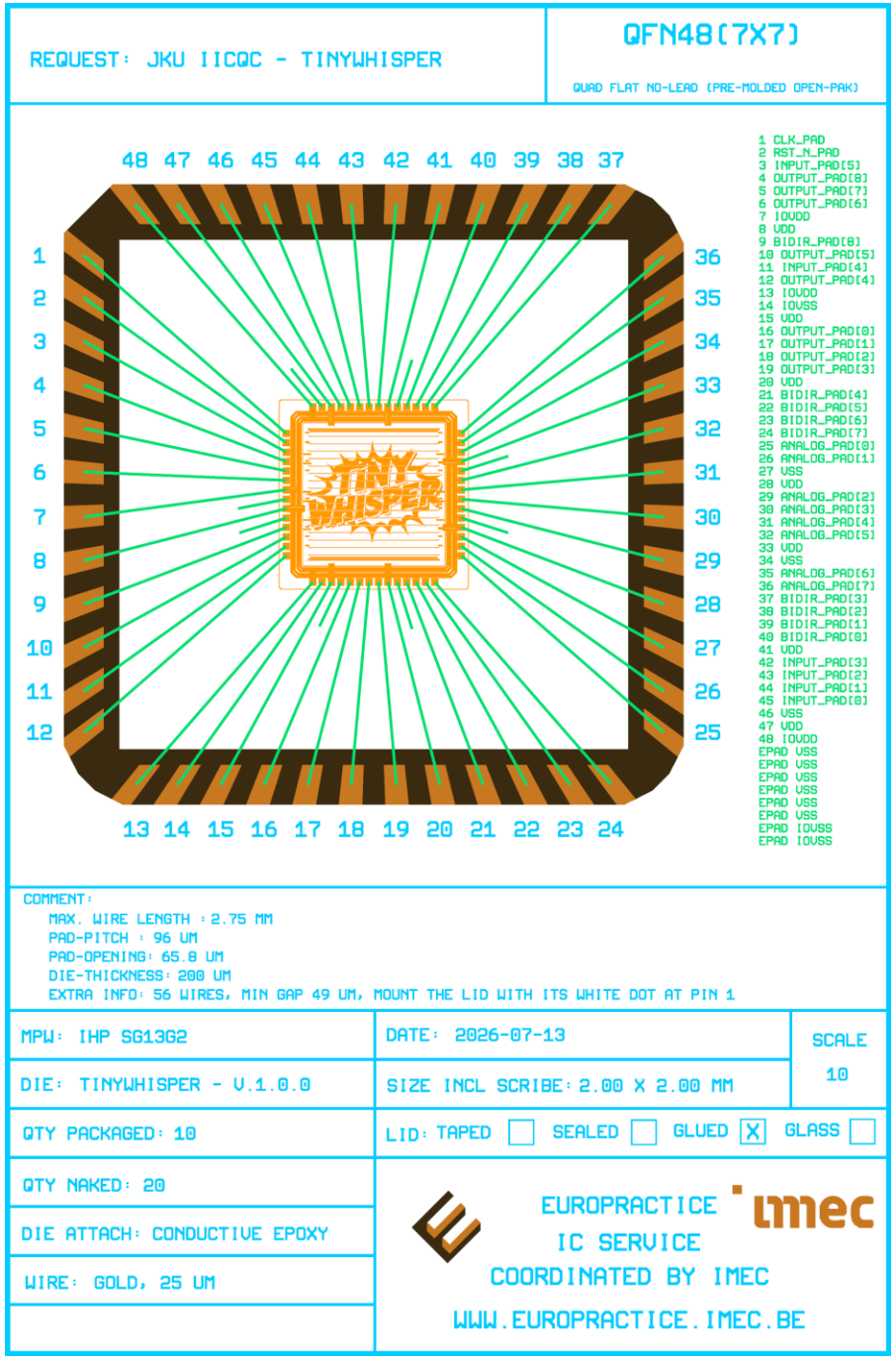


IQ Modulator



Bonding Diagram

- Automatic generation:
  - Python script
  - Config file (pinout, placement)
  - Extract layers (TM2, pads)
  - Place chip layout
  - Get pad centers
  - Draw bond wires
  - Render GDS



Tools Installation



Local Installation

1. Install **VS Code**: https://code.visualstudio.com
2. Run the **installer** (Git, Docker, WSL2 / XQuartz, IIC-OSIC-TOOLS):
   - Windows (admin PowerShell): `irm https://osic.tools/install.ps1 | iex`
   - macOS / Linux: `curl -fsSL https://osic.tools/install.sh | bash`
3. Reboot, then start Docker Desktop once
4. Start the container in the `iic-osic-tools` folder:
   - Windows: `.\start_vnc.bat` macOS / Linux: `./start_vnc.sh (or start_x)`
   - Open http://localhost/?password=abc123 (first start pulls ~4 GB)
5. Your designs: `~/eda/designs` (laptop) = `/foss/designs` (container)

Containerized On-Demand Workspace System (COWS)

1. Open COWS: https://jku128011.riic.uni-linz.ac.at/
2. Log in and change password
3. Create a workspace
    a. Go to “Workspaces” and click “Create workspace”
    b. Enter “Workspace name” and select “KV VLSI Design 26/27” template
4. Start the workspace → “Status” should change to “Running”
5. Open it by clicking the workspace name
6. **Enjoy your workspace! Happy designing!**



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